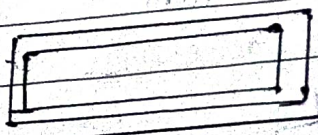
★ Symbols used in ER Model :-

Figures	Symbols	Represents
1) Rectangle		Entities in ER Model
2) Ellipse		Attributes in ER Model
3) Diamond		Relationships among entities
4) Line		Attributes to entities & Entities to other relationship types
5) Double Ellipse		Multi-valued Attributes
6) Double Rectangles		Weak entity.

★ Entity :-

- It represents a real world object, concept or thing about which data is stored in database.
- It acts as building block of database.
- Tables in relational database represent these entities.

Ex:- Real world objects :- Person, car, Employee etc.
Concepts :- Course, Event, Reservation etc.
Things :- product, Document, Device etc.

★ Types of Entity :-

1) Strong Entity :-

- A type of entity that has a key attribute that can uniquely identify each instance of entity.
- A strong entity does not depend on any other entity in the schema for its identification.

- It has a primary key that ensures its uniqueness and is represented by a rectangle in ER diagram.

2) Weak entity :-
- It can't be uniquely identified by its own attributes alone.
- It depends on a strong entity to be identified.
- It is represented by double rectangle.
- The relationship between strong entity & weak entity is called as identifying relationship and is represented by double diamond.

★ Attributes in ER Model :-
- Attributes are the properties that define the entity type.
- For ex, for student entity, Roll-No, Name, DOB, Age, Address and Mobile-No are attributes.

Types of Attributes :-

1) Key Attributes :-
- The attribute which uniquely identifies each entity in entity set is called the key attribute.
- For ex :- Roll-No will be unique for each student.
- In ER diagram key attribute is represented by an oval with an underline.

2) Composite Attribute :-

- An attribute composed of many other attributes.
- For ex, address attribute consits of street, city, state & country
- It is represented by an oval comprising of ovals.

3) Multivalued Attribute :-
- An attribute consisting of more than one value for given entity
- Ex :- Phone_no (can be more than one for a student)

4) Derived Attribute :-
- An attribute that can be derived from other attributes.
- Ex :- Age (can be derived from DOB).

★ Relationship Type :-
- The relationship type represents association between entity types.
- Ex :- "Enrolled in" is a relationship type that exists between student & course.

★ Relationship Set :-
- A set of relationship of same type is called as relationship set.

★ Degree of a Relationship Set :-

1) Unary / Recursive Relationship :-
When there is only one entity set participating in a relationship.
Ex :- one person married to only one person.

2) Binary Relationship :-
- When there are two entity sets participating in a relationship.



3) Ternary Relationship :-
- When there are three entity sets. participating in a relationship